



Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

Participant Name: Shruti Patel

Hackathon Track: AI for Better Living and Smarter Communities (Healthcare Access & Community Wellness)

Project Title: AyuGuard — Ambient Caregiver Platform for Independent Elderly Living

Brief about the idea

The Core Healthcare Challenge for Seniors

- Independent Living Risks: Millions of seniors live alone. Gradual health declines (dehydration, respiratory infections, cognitive change) are subtle and frequently ignored by the senior themselves, leading to late-stage emergency hospitalizations.
- The Caregiving Gap: Busy family caregivers lack continuous visibility. Passive wearables only capture simple raw vitals, while home camera monitoring is highly intrusive and violates elderly dignity.

AyuGuard's Ambient Solution

- Conversational Care: AyuGuard allows seniors to simply type about their day. Using Google ADK and Gemini, it automatically extracts clinical symptoms from casual stories.
- Non-Intrusive Safety: Seamlessly maps logs, cross-references symptoms with database tables, scores urgency, and generates warm, stress-reducing updates for caregivers, keeping independent seniors safe without intrusive camera monitoring.

How We Approached the Problem Statement

- **Multimodal Data Ingestion:** Accepts unstructured text senior chats, along with uploaded medical records (PDF prescription scans, blood panels) directly into Google Cloud Storage.
- **Modular Multi-Agent Architecture:** Built using the Google Agent Development Kit (ADK 2.x). Instead of one generic LLM, AyuGuard separates responsibilities into specialized sub-agents: one for clinical extraction, one for care plan drafting, and one for caregiver outreach.
- **Zero-Hallucination Rigor:** Matches extracted symptom phrases against a reference clinical symptoms dataset (40+ disease-symptom clusters) using a python-based Jaccard similarity scoring formula to ground all AI reasoning.

Real-World Practical Impact

- **Proactive Care Triage:** Evaluates frequency, severity, and longitudinal trends over a multi-day sliding window to output a clear, deterministic urgency score (Green, Amber, Red). This triggers early clinical outreach, preventing emergency room crises.

VS Differentiation

- **Passive Wearables** → Track vitals but miss qualitative symptoms; AyuGuard understands natural language.
- **Intrusive Home Cameras** → Cause stress and privacy issues; AyuGuard is ambient, conversational, and dignity-preserving.
- **Standard Medical Chatbots** → Give generic replies without clinical grounding; AyuGuard is strictly tied to medical reference tables.

★ Unique Selling Proposition (USP)

- **Hybrid Intelligence Engine** → Combines Gemini conversational AI with deterministic clinical rules for reliable outcomes.
- **Dual-Persona Messaging** → Splits logs into clinical care plans for health workers and empathetic updates for caregivers.

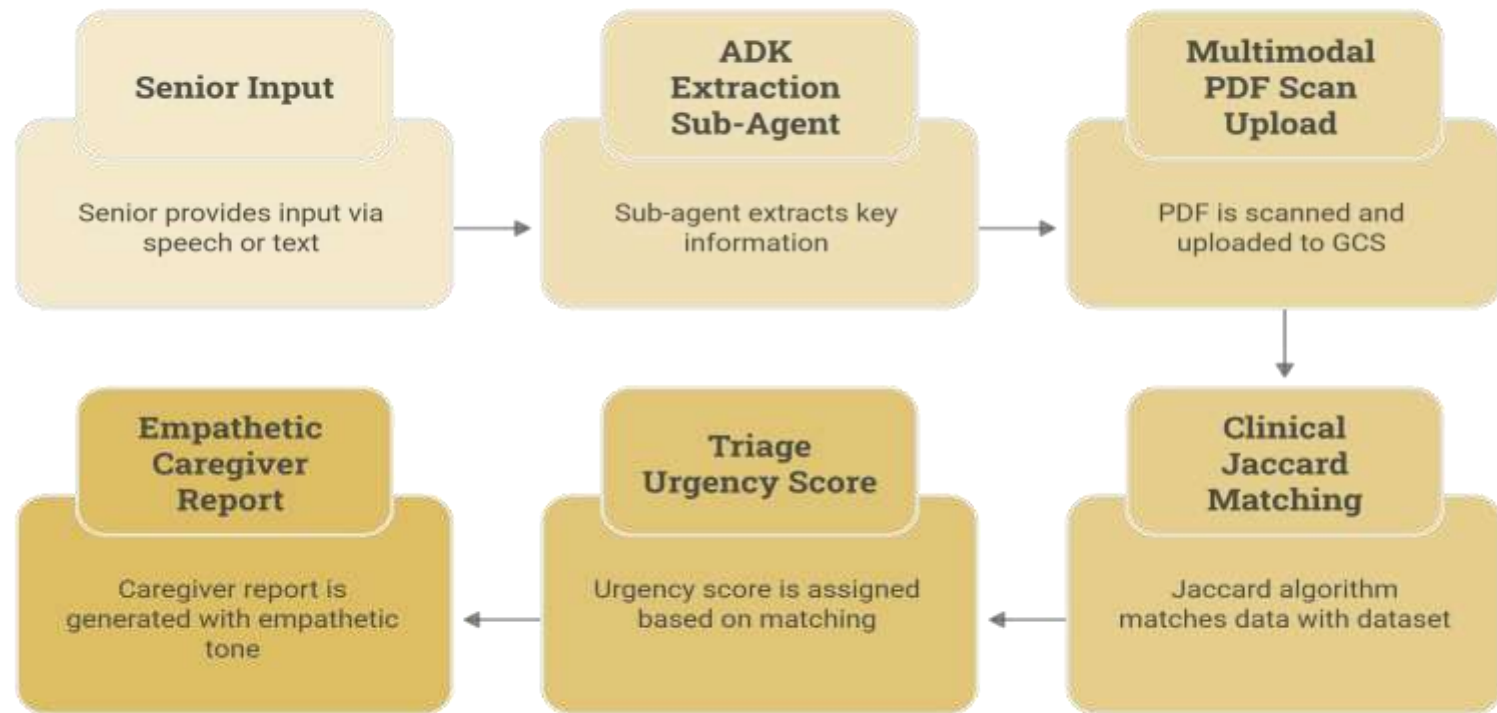


Core Intelligent Features

Feature	Description	Technology	Benefit
 Text Logging	Captures symptoms from natural conversation	NLP Parser	Enhances accuracy, reduces manual entry
 Dynamic Jaccard Symptom Mapper	Links casual symptom expressions to disease profiles	Jaccard Index	Nuanced understanding, appropriate diagnoses
 Multimodal PDF Record Vault	Parses uploaded medical reports	Gemini Vision	Streamlines integration, enhances continuity of care
 Longitudinal Trend Window	Tracks symptom occurrence over time	Multi-day sliding window	Informed decisions, proactive treatment adjustments
 Deterministic Urgency Scorer	Assigns triage level based on metrics	Frequency, severity, duration	Efficient triage, timely attention
 Dual-Persona Digest Dispatcher	Generates care task-list and family digest	Structured care task-list	Bridges communication gap, fosters supportive environment

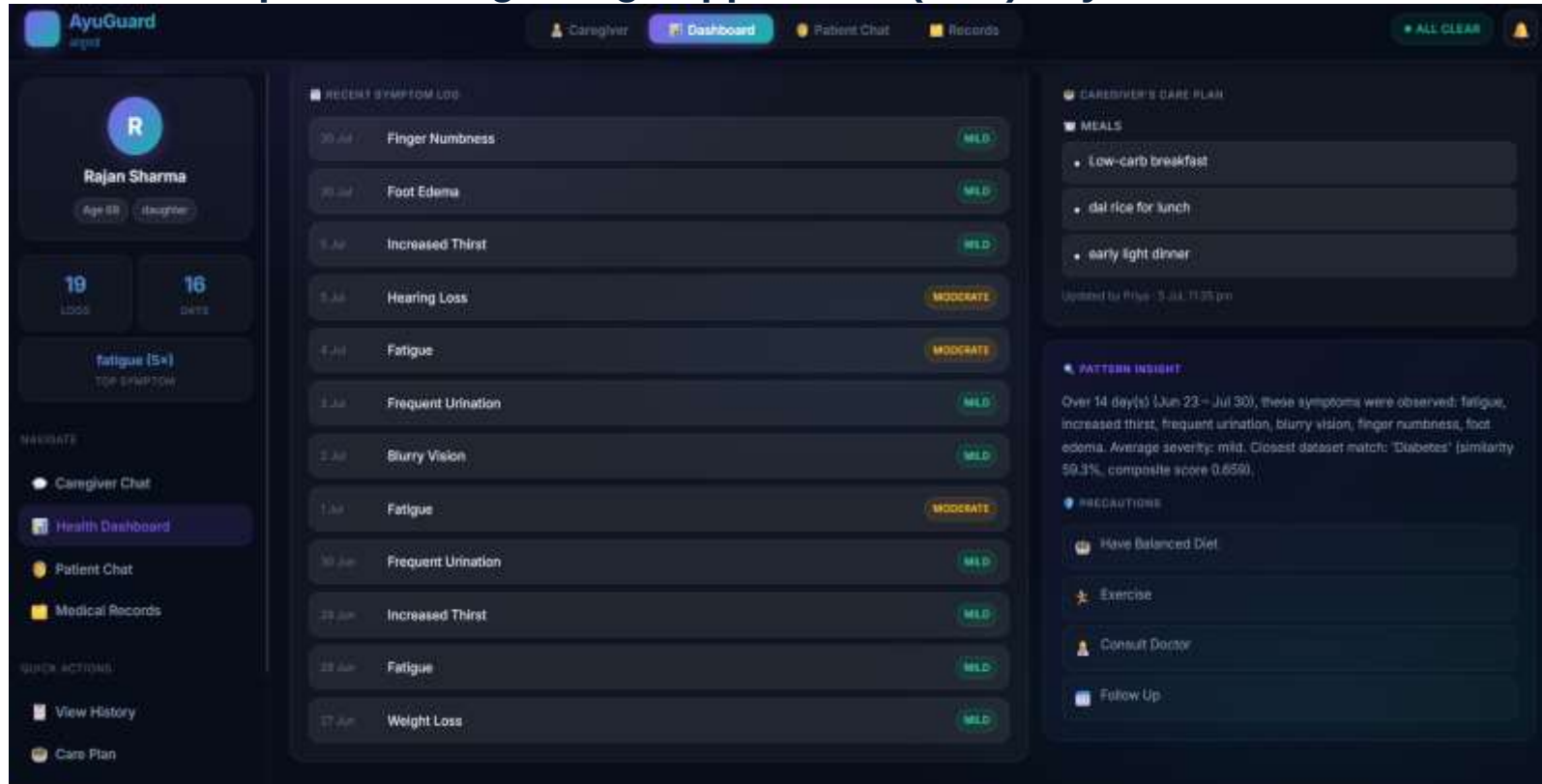
- **Text Logging** → NLP parser captures conversational symptoms.
- **Jaccard Symptom Mapper** → Links casual expressions to disease profiles.
- **PDF Record Vault** → Gemini Vision parses medical reports into profiles.
- **Trend Window** → Tracks symptom frequency over time.
- **Urgency Scorer** → Triage levels (Green/Amber/Red) based on metrics.
- **Digest Dispatcher** → Care task-list for nurses + family digest.

End-to-End Decision Intelligence Workflow

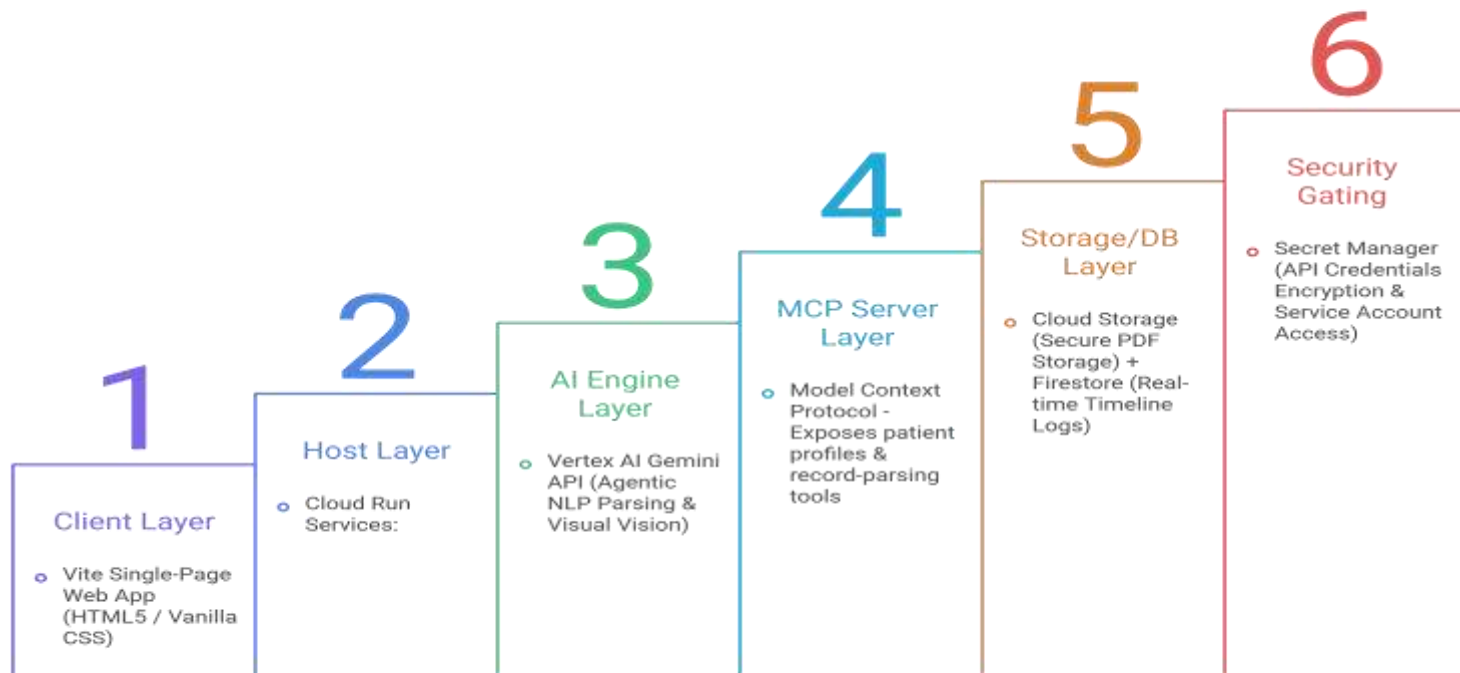


Wireframes/Mock diagrams of the proposed solution

Responsive Single-Page Application (SPA) Layout



Serverless Google Cloud System Architecture



Made with Napkin

☞ Live SPA Dashboard URL: [Live Demo](#) ☞ Live API Orchestrator URL: [Orchestrator Link](#)

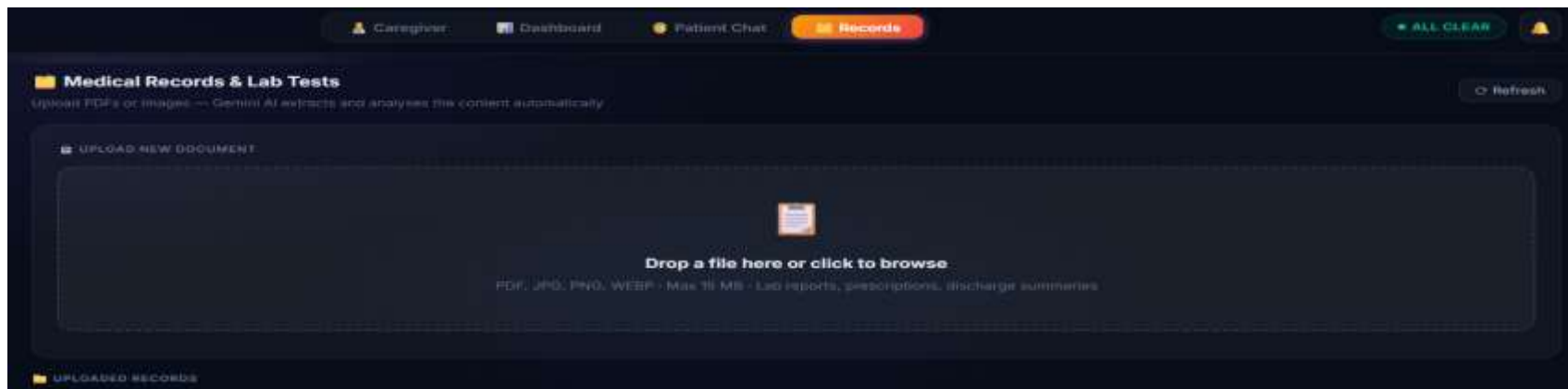
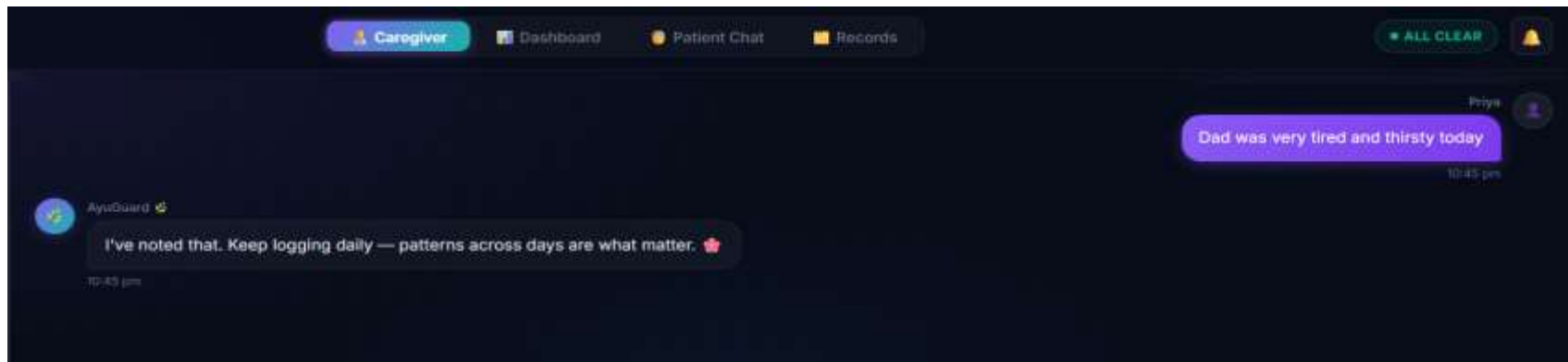
Technologies used in the solution :

- **ADK 2.x** → Modular Python framework for specialized sub-agents with strict boundaries.
- **Vertex AI (Gemini)** → Advanced NLP + multimodal extraction for unstructured entries and scanned medical PDFs.
- **Cloud Run** → Serverless hosting with auto-scaling and zero idle cost.
- **GCS & Firestore** → Secure patient PDF storage + real-time caregiver feed syncing.
- **Secret Manager** → Encrypted credential handling with enterprise-grade compliance.

Google Cloud & ADK AI Stack Components



Snapshots :



Snapshots:

Caregiver Dashboard Patient Chat Records

ALL CLEAR

Patient Mode — Share how you're feeling, ask health questions, or view care plan updates from your caregiver.

Rajan Sharma

I am having bodu aches

11:04 pm

AyuGuard 🌿

Pranaam Rajan ji,

I hear you're having some body aches. I've noted this down for you. Priya can also see this on her dashboard.

Please try to get some rest. Since the symptom pattern resembles something that needs a doctor's advice, I would strongly recommend you let Priya know if you feel worse, and please consult a doctor for proper evaluation and treatment.

I hear you're having some body aches. I've noted this down for you. Priya can also see this on her dashboard.

Please try to get some rest. Since the symptom pattern resembles something that needs a doctor's advice, I would strongly recommend you let Priya know if you feel worse, and please consult a doctor for proper evaluation and treatment.

11:05 pm

Google Cloud 

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Thank you

